App Dev I Project Report

1. Student Details

Name: Iqra Mohammedi Afif
Roll Number: 21f3000425
Email: 21f3000425@ds.study.iitm.ac.in
About Me: I am currently in diploma level of IITM BS Degree, simultaneously pursuing my Masters in Statistics (final semester). I was also working as an Online Math tutor (resigned in 2025) and an offline Math tutor (resigned recently). I enjoy learning and exploring statistics and understanding statistics as foundation of data science.

---

2. Project Details

Project Title: Placement Portal App

Problem Statement:

To design and build a web-based application that manages campus recruitment activities involving companies, students, and placement drives. And be able to approval companies, track student applications, avoid duplicate registrations, and maintain placement records.

Approach:

The app was built using Flask for application back-end. Jinja2 templating, HTML, CSS and Bootstrap for application front-end, and SQLite for database. It allows companies and students to register, create placement drives, allows students to apply and keeps a track of all these activities

---

3. AI/LLM Declaration

I used ChatGPT as a guide, to understand and improve. I used it to assist me in fixing bugs, explaining logics, suggestions to improve the structure and few code suggestions for backend help. The extent of AI/LLM usage is around 25–30%.

---

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap	Frontend styling and responsive design
SQLite	Lightweight local database for storing user data
CSS	Styling of webpages
HTML	Structure of webpages

---

5. Database Schema / ER Diagram

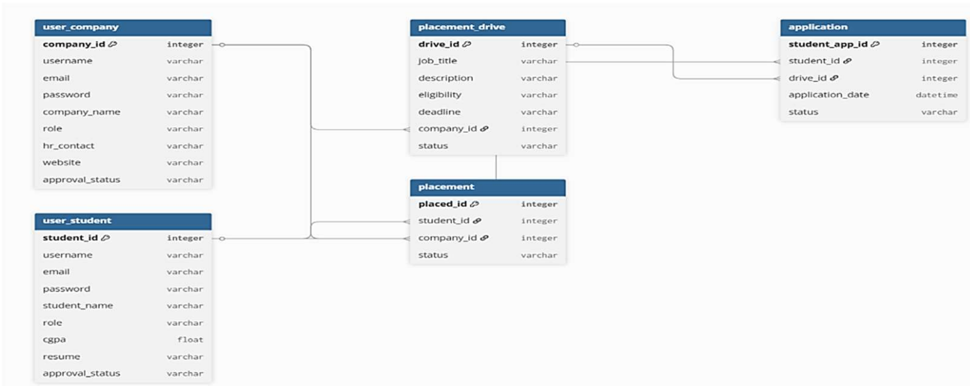
Tables:

1. User_company – stores company details.
2. User_student – store student details.

3. Placement drive – stores jobs created by company.
4. Application - stores about the application to jobs by student.
5. Placement - stores the students who got placed.

Relationships:

- One company can create multiple placement drives (one to many)
- One placement drive can have multiple applications (one to many)
- One student can apply to multiple drives (one to many)



---

6. API

Endpoint	Method	Description
/login	POST	Authenticate user
/register	POST	Register student or company
/company	GET	Display company dashboard
/student	GET	Return daily/weekly analytics summary

And many other endpoints are used and is shown in yaml file.

YAML API Definition File:

Included separately in the submission ZIP as api.yaml.

---

7. Architecture and Features

Architecture Overview:

The main application logic is implemented in the app.py file, which acts as the controller handling all routes and user interactions. The database is managed using SQLAlchemy ORM with SQLite as the backend, where models define the data structure. The frontend is developed using HTML templates rendered through Jinja2, allowing dynamic content generation. These templates are stored in the template's directory, while CSS is stored in the static directory. Session management is handled using Flask sessions to maintain user authentication and access control.

Implemented Features:

All core features are implemented to the best of my knowledge.

---

8. Video Presentation

Drive Link:

https://drive.google.com/file/d/17DIVI7eBygwSGAwEuua6531Ie8W-7gYv/view?usp=sharing